

answer, it will not matter much as it is not critical. But think of a situation where a computer vision system outputs wrong information in skin cancer detection, or a self-driving car is unable to detect an object in front of it, the situation where the visual object detection system fails to detect the enemy in a military operation?

The results would be catastrophic and cause irreparable damage. These are critical applications where accuracy, timing and quality are extremely important.

AI technology related to computer vision, image processing, machine learning and deep learning has many applications across industries.

In consumer electronics, these applications provide face recognition and autofocus to camera-enabled devices.

In the public sector, these can be used to identify dangerous driving situations with traffic monitoring cameras.

In healthcare, these can be used for medical imaging and diagnosis.

In retail, these can be used to spot malicious activities in stores.

In agriculture, these can be used to determine the health of crops in the field.

All of these applications are critical to core businesses and processes.

In critical applications, quality and quantity of input data are important to obtain correct results. Deep learning based on a deep neural network, a class of machine learning algorithm, is important in mission-critical applications. Deep learning uses multiple layers to progressively extract higher-level features from raw data input, enhancing accuracy and giving more reliable results.

Key differences in mission-critical AI

The context and levels of criticality may differ from industry to industry, but there are a few parameters common to all such applications involving AI technology. **Mission-critical, time-critical, life-critical and safety-critical applications require fast identification, timely action, data delivery, reliable notification and warning messages.** Key differences in mission-critical AI applications

are accuracy, service-level agreement (SLA), scalability and security.

Accuracy. Mission-critical AI applications need significant and better accuracy. This means that training data needs to be greater in quality and quantity. Actually, quantity of data you need depends on complexity of algorithms for a particular problem.

SLA. All core business processes have strict SLAs. An SLA is a commitment between a service provider and a client. The commitment includes quality, availability and responsibilities. Services should be provided to the customer as agreed upon in the contract. For example, when an image is fed to a model for prediction, as per requirement, the model should not take much time to predict the correct result. If there is a delay, it is not following a strict SLA.

SLAs may include many components, from a definition of services to termination of agreement.

Scalability. Besides SLAs, architecture has to be scalable. High accuracy and strict SLAs need an architecture that can effortlessly scale for both computing and storage.

Security. This is one of the prime concerns in AI technology. Certain applications present security and data privacy concerns. Data may include sensitive information that has to be stored securely with enterprise-grade security features. In mission-critical AI applications, strict security measures are taken into account and only authorised users have access to information.

AI components in mission-critical applications

Essential components for mission-critical AI applications include datasets and hardware setup. In machine learning, construction of algorithms is done through building a mathematical model from input data.

Datasets. Quantity and quality of data are important when it comes to AI applications. Data used to build the final model usually comes from multiple datasets. In most cases, three datasets are used in different stages of the creation of the model. For any given dataset, there are many possible models to identify them with fundamentally different interpretations.

Input data. This is one the most



Screenshot of Not Hotdog app (Credit:www.oreilly.com)

important components when it comes to AI technology. Imagine there are many cameras installed in a retail store having different outlets. From input data received from these camera sensors, the user may have to process thousands of images per minute to detect malicious activities.

Training data. For mission-critical AI applications, accuracy and quality are important. An AI model is useful if input data is trained on. For example, implications of failing to recognise skin cancer can be fatal. We need datasets that are significantly greater in quantity and quality. This means we may have to spend significant time in analysing, preparing and labelling data.

Feedback. A feedback loop improves the model. For example, in the retail store, it would be a good idea to label false predictions and feed them back to the model for retraining. A continuous feedback loop increases training data size, and compute requirements also increase but with better accuracy and results.

Hardware setup. AI is based on algorithms, but without a hardware setup or platform, algorithms do not make any sense. **Unfortunately, the importance of hardware is not discussed or included in most of the materials found on the Internet.**

An AI ecosystem contains various types of agents including hardware and software. Hardware includes sensors, processors and storage. Hardware used for running mission-critical applications also depend on SLAs and differences in data set sizes.

Sensors. These detect or measure physical properties, and record, indi-